Hamza Naeem

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Education

Northeastern University, Boston, MA — Sep 2024 – Apr 2026
Master of Science in Robotics, GPA: 4.00/4.00 (all A grades), Fulbright Scholar
Courses: Autonomous Field Robotics, Mobile Robotics, Robot Sensing & Navigation, Advanced Computer Vision, Machine Learning & Pattern Recognition

National University of Sciences & Technology (NUST), Islamabad, Pakistan — Sep 2017 – Jul 2021
Bachelor of Electrical Engineering, GPA: 3.74/4.00, Top 5%
Courses: Computer Vision, Digital Image Processing, Digital Signal Processing, Linear Control Systems, Machine Learning

Research Interests

Field robotics • Robot perception • 3D reconstruction and mapping (SfM, SLAM) • Uncertainty-aware perception • Marine robotics • Change detection across repeat surveys • Visual-inertial navigation and sensor fusion • Machine learning • Robot manipulation

Publications

Peer-Reviewed

- **Hamza Naeem**, Dennis Giaya, Hanumant Singh. "Feature Matching for Ocean Robotics: A Structure-from-Motion Benchmark Across Marine, Polar, and Aerial Datasets". *IEEE OCEANS 2026*

Workshop Papers

- **Hamza Naeem**. "Reads Right, Judges Wrong: Near-Miss Failures and the Limits of Uncertainty Recovery in VLM Success Detectors". Accepted, *CoRL 2026 Workshop on Modeling Uncertainty in Robotic World Models (UncertainWM)*
- **Hamza Naeem**. "Pixels Without Geometry? Auditing the 3D Consistency of Video World Models with Structure-from-Motion". Accepted, *CoRL 2026 Workshop: Do Robots Need World Models?*
- **Hamza Naeem**. "Lower Means Closer? Layout Bias in VLM Spatial Judgments on Underwater and Off-Road Imagery". Under review, *CoRL 2026 Workshop on Spatial Intelligence and Structured Reasoning for Embodied Navigation (SISREN)*

Research Experience

Graduate Research Assistant, Field Robotics Lab (Prof. Hanumant Singh) — Sep 2025 – Present
Northeastern University, Boston, MA — (remote since May 2026)

- **Feature-Matching Benchmark for Ocean Robotics (IEEE OCEANS 2026):** Benchmarked nine feature-matching pipelines (SIFT to RoMa) in one COLMAP SfM framework on a deep-sea survey of the Bio9 hydrothermal vent (HOV Alvin, $\sim$2,000 m), a Svalbard glacier UAV pass, and a heritage-building control; showed that registration completeness overstates reconstruction quality: RoMa registers all 108 vent images, yet 28% of its adjacent camera pairs disagree with the cross-method consensus, while ALIKED registers 99 with none in 27 min vs. 9.1 h (code: github.com/neufieldrobotics/ocean-sfm-benchmark)
- **Underwater Photomosaicing of a Roman Shipwreck:** Mosaicked the Skerki Bank site from 267 low-contrast seafloor images (SIFT and SuperPoint + SuperGlue matching, RANSAC, Levenberg-Marquardt homography refinement), optimizing image poses in a GTSAM factor graph; loop closures corrected drift on 6- and 29-image subsets, and SIFT outperformed SuperGlue on the larger image sets
- **UAV-Based Polar Glacier Change Detection & Alignment Study:** On consecutive drone surveys of a marine-terminating Svalbard glacier, tested how much alignment matters for change estimates (joint vs. separate COLMAP reconstructions, GPS vs. TEASER++/GICP registration on stable terrain, M3C2 vs. DEM volumes): the same flights gave -9 to $-1{,}762\,\mathrm{m}^3$, and a 0.25 m vertical bias alone would add $\sim$2,250 m^3 of apparent change
- **3D Reconstruction & Change Detection for Quasi-Dynamic Scenes:** Built a COLMAP-based pipeline that reconstructs and segments scenes containing articulated, multi-state objects, where the static-world assumption of conventional SfM breaks down; registered per-state point clouds via FPFH + RANSAC and separated moving components from static background geometry to recover which objects changed state

- **Real-Time Targetless Camera-LiDAR Calibration:** With a four-person team, developed an online extrinsic calibration pipeline aligning a 3D LiDAR (Ouster OS1) and a camera (FLIR Blackfly S) without checkerboards, matching LiDAR planes to 2D semantic edges from the Segment Anything Model (SAM) via tiered translation/rotation optimization; tested by injecting known extrinsic offsets into NeURoam sequences

Independent Research, World Models and Robot-Learning Evaluation Fall 2026

- **Geometric Consistency of Video World Models:** On real DROID wrist-camera clips, scored three image-to-video models (LTX-Video, CogVideoX, Wan2.2) and the action-conditioned Ctrl-World with COLMAP, epipolar geometry, and VGGT against controls built from real footage; showed that a frozen frame beats real footage on raw geometric scores, and distilled a reporting protocol based on motion-matched real references
- **Near-Miss Failures in VLM Success Detectors:** Built near-miss benchmarks (rendered tabletop arrangements with exact ground truth, and real TACO-Play episodes with counterfactual instructions) and showed that VLM success detectors (Qwen3.5, Gemma 4) read failed outcomes correctly yet still pass them; conformal thresholds recover the strongest model on controlled scenes but not on real robot frames
- **Failure Probes for a Robot Foundation Model** (github.com/naeem-ha/vla-failure-probe): Built a $\pi_{0.5}$ + LIBERO probe that switches a robot's instruction mid-task and logs every step; told to "stop" before grasping, the policy ignored it in 6 of 6 simulated runs and picked up the bowl anyway

Professional Experience

SVP, Robotics & AI, Cowlar, Inc. (Y Combinator W17) May 2026 – Present

- Lead 35+ engineers across 7+ robotics, machine-learning, and automation projects for North American manufacturing clients
- Fine-tuned a Qwen3-VL vision-language model (LoRA) for fiber inspection on the Fiber Termination and Fiber Sorting robots

Robotics and AI Intern, viaPhoton, Inc., Naperville, IL May 2025 – Aug 2025

- Built vision-based inspection and automation for fiber-optic assembly lines on the factory floor, spanning industrial camera selection and integration, detection-model development, and deployment onto edge compute at the line

Principal Design Engineer ← Team Lead, Industrial Automation ← Computer Vision Engineer Aug 2021 – Sep 2024
Cowlar, Inc. (Y Combinator W17), Islamabad, Pakistan

- Built a Fiber Termination Robot that combines YOLO-based vision models with imaging algorithms (masking, template matching) for visual inspection, alignment, and insertion of 250 μm optical fibers, achieving a 97% insertion success rate with 5 μm motion resolution through real-time parallel control of vision-based automation
- Owned the actuation and real-time motion-control stack across the Fiber Aligner and Fiber Termination robots: Festo servo and pneumatic actuators (CANopen/CMMT, Modbus/CTEU), Oriental and Faulhaber motor drives, and multi-protocol integration, closing the loop on vision feedback to cut cycle time by 74% under safety-critical constraints
- Built the computer vision, ML, and hardware-integration software behind Fiber Organizer, a fiber-sorting system patented by viaPhoton (US Patent 12,001,073 B2), achieving 99% sorting accuracy and cutting sorting time from 180 s to 8 s; it now runs $\sim$10,000 sorts daily with a 0.1% model-related sorting failure rate
- Integrated Time-of-Flight depth sensors with a Jetson Nano in C++, fusing 3D depth with RGB to make activity detection robust to monocular ambiguity
- Built a GPS-denied tracking and localization system for factory material trolleys on Raspberry Pi 4B, fusing two monocular cameras with an IMU and edge detection, cutting production downtime by $\sim$30%
- Deployed quantized detection models across Raspberry Pi, Jetson, and RK3588 NPU hardware, cutting inference latency by up to 60%

Teaching Experience

Teaching Assistant, ME 5250 Robot Mechanics & Control (Prof. Bahram Shafai) Jan 2025 – May 2025
Department of Electrical & Computer Engineering, Northeastern University

- Held office hours and graded coursework for 82 graduate students on kinematics, dynamics, motion planning, and control, with MATLAB applications; helped students solve problems in class when the professor was away

Academic Service

Reviewer, CoRL 2026 Workshops: *Do Robots Need World Models?* and *Spatial Intelligence and Structured Reasoning for Embodied Navigation (SISREN)* 2026

Selected Projects

Hardware-in-the-Loop Sensor Emulation for Robot Testing Jul – Aug 2026

- Wrote a C++17 daemon that impersonates a Livox Mid-360 LiDAR on the wire (200,000 pts/s point and 200 Hz IMU streams), so the stock, unmodified `livox_ros_driver2` publishes point clouds from replayed or synthetic data; validated against the real ROS 2 Humble driver in a QEMU aarch64 guest
- Built a companion harness presenting synthetic or replayed IMU and Intel RealSense streams over CycloneDDS, indistinguishable from live hardware to standard consumers, with reproducible fault injection for degraded-sensor testing

Multi-Modal SLAM Benchmark: Visual, RGB-D & LiDAR (ROS 2), Northeastern University Feb – Apr 2025

- In a four-person team project for Robotics Sensing & Navigation, built and benchmarked SLAM across three sensor modalities: monocular ORB-SLAM3 on EuRoC and on live runs with a calibrated Logitech C920, RGB-D ORB-SLAM3 on the Kinect-recorded TUM RGB-D dataset, and a ROS 2 LiDAR SLAM pipeline run live on an LDS-01 scanner with NDT/GICP scan matching, submaps, and g2o pose-graph optimization
- Compared the modalities on texture dependence, depth accuracy, lighting robustness, motion sensitivity, and map quality, showing where visual SLAM loses tracking (textureless walls, darkness) and LiDAR does not

GPS-Denied Vehicle Navigation via IMU Fusion & Dead Reckoning, Northeastern University Jan – Apr 2025

- Built a dead-reckoning pipeline estimating a vehicle's 2D trajectory from a VectorNav VN-100 IMU, benchmarked against GPS over a multi-km drive; calibrated the magnetometer for hard- and soft-iron distortion via eigenvalue decomposition and fused gyroscope and magnetometer yaw with a tuned complementary filter
- Corrected accelerometer-derived velocity using GPS stationary periods, polynomial drift removal, and piecewise scaling, and quantified how long inertial-only navigation stays trustworthy

Pick-and-Place Control for a PincherX-100 Manipulator, Northeastern University Oct – Dec 2024

- Programmed a perception-driven PincherX-100 arm in MATLAB and C++: overhead-camera calibration and 3D geometry-based pose estimation, homogeneous transformations, and inverse kinematics for obstacle-aware trajectory planning and execution

Awards, Honors & Scholarships

- **Fulbright Scholar**, US Department of State (2024): selected from 10,000+ applicants; fully funded MS through the Fulbright Award, Dean's Graduate Scholarship, and Global Learners Award
- **1st Place**, Draper University × YC AI Startup School Hackathon (2025) • **Best AI for Longevity Innovation**, Longevity, AI & Cognitive Hackathon at MIT Media Lab (2024)
- **Houston Award** (3×), Cowlar's company-wide engineering award (2022–2023) • **Merit Scholarship**, NUST, four consecutive semesters (cohort's top five); Dean's Honor List (2021)

Technical Skills

Programming: Python, C++17, Rust, C, MATLAB, Bash, CMake
3D Vision & Perception: Structure from Motion (COLMAP), VGGT, multi-view and epipolar geometry, bundle adjustment, point cloud registration (ICP, GICP, FPFH, TEASER++), SLAM (ORB-SLAM3, GTSAM, g2o), extrinsic calibration, sensor fusion
Robot Learning & ML: PyTorch, vision-language-action policies ($\pi_{0.5}$), vision-language models (Qwen3-VL, LoRA fine-tuning), diffusion models, detection, segmentation, pose estimation, conformal calibration
Robotics & Simulation: LIBERO (MuJoCo), Isaac Sim, AirSim, ROS 2, hardware-in-the-loop testing, forward/inverse kinematics, motion planning, servo and pneumatic motion control (CANopen, Modbus), NVIDIA Jetson, RK3588